Distributed Systems

5th Lecture

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Outline

- Communication
  - Layered Protocols
  - Middleware Protocols
  - Types of Communication
- Communication Methods
- Q&A

Message Passing Requirements

- Agreements Needed at a Variety of Levels
  - Meaning of the Bits Being Sent
    - Character coding: e.g., EBCDIC and ASCII
  - Number of Volts for a 1-bit
  - Indication of the Last Bit of the Message
  - Detection of Damaged or Lost Messages
  - Lengths of Numbers, Strings, and Others
  - Representations

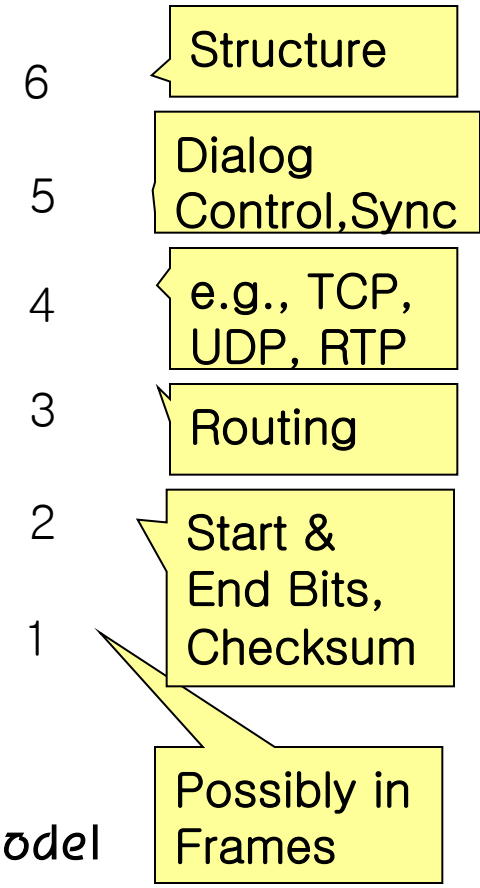
Agreements from the low-level details of bit transmission to the high-level details of how information is to be expressed

Layered Protocols

- ISO OSI (Open Systems Interconnection) Reference Model
  - Designed to Allow Open Systems to Communicate
    - Open system is prepared to communicate with any other by using standard rules that govern the format, contents, and meaning of messages
      - Protocols: such rules formalized
        - Connection oriented
        - Connectionless
  - Useful for Understanding Computer Networks
- Protocol Suite (or Stack)
  - Collection of Protocols Used in a System

Illustration: Layered Protocols (1)

Figure 4-1 on Page 118



Layers, interfaces, and protocols in the OSI model

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Illustration: Layered Protocols (2)

Figure 4-2 on Page 118

A typical message as it appears on the network

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Middleware Protocols

Figure 4-3 on Page 123

An adapted reference model for networked communication

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Types of Communication

Figure 4-4 on Page 124

Viewing middleware as an intermediate (distributed) service in application-level communication

Persistent vs. Transient Communication
Asynchronous vs. Synchronous Communication

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Communication Methods

- RPC (Remote Procedure Call)
  - Communication by Calling Remote Procedures
    - Definition of service interface
    - Lack of ability to create new object instances
    - Lack of support for remote object references
- RMI (Remote Method Invocation)
  - Communication by Calling Methods of a Remote Object
    - Implementation of a remote interface
    - Creation of new object instances
    - Support for remote object references

Communication Methods (Cont'd)

- Socket
  - Communication of Messages and Data between Processes
    - Use of a raw communication channel
    - Definition of a low-level message protocol
    - Definition of data transmission format
- Distributed Event-Based Systems
  - Communication via Event Subscription and Notification
    - Support for heterogeneity
    - Support for asynchronous communication

Middleware Approaches

- Location Transparency

Figure 2.7 on Page 68

Method Invocation

- Local vs Remote Invocation

Figure 5.12 on Page 223

Distributed Object Model

- A Remote Object and Its Remote Interface

Figure 5.13 on Page 224

Distributed Object Model (Cont'd)

- Instantiation of Remote Objects

Figure 5.14 on Page 225

RMI Components

- Proxy
  - Forwarding Messages to a Remote Object and Receiving the Reply
    - Making RMI transparent to clients
- Dispatcher
  - Receiving the Request and Selecting the Appropriate Skeleton Method
- Skeleton
  - Implementing Methods in the Remote Interface
    - Unmarshalling arguments and invoking the method

RMI Components (Cont'd)

- Illustration

Figure 5.15 on Page 226

Remote Reference Module

Translating between Local and Remote Object References and Creating Remote Object References

RPC Components

- Illustration

Figure 5.10 on Page 217